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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: B.Tech

Year: II Semester:III

Stream/s ;CSBS

Subject: Object Oriented Programming

Time: 3 hrs (10:00am to 1:00pm)

Date: 28 / 11 / 2023

No. of Pages: 03

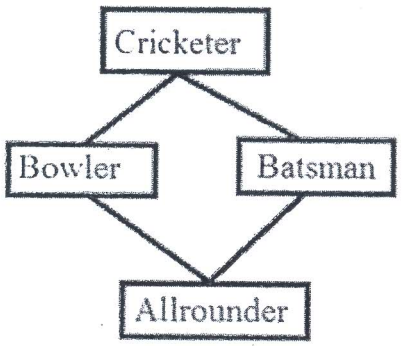
Marks: 100

Final Examination (2023-24) / Re exam (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	[20]
CO-1; SO-1; BL-1	a.	Define relational operator and with an example.	[05]
CO-2; SO-1; BL-6	b.	Write Advantages and Disadvantages of Object Oriented Programming	[05]
CO-3; SO-1; BL-4	c.	Difference between Data Hiding and Encapsulation (five points)	[05]
CO-4; SO-2; BL-1	d.	Define file handling use and write about the fstream.h header file with the help of diagram.	[05]
Q2 CO-3; SO-1; BL-1,6	a)	Define function template. Write a C++ Program to find Square function of int, float, double data-type using single template.	[10]
	b)	Write a C++ program to create a class called Car that has private member variables for company, model, and year. Implement member functions to input and output these variables.(Hint: scope resolution operator)	[10]

Q3 CO-2; SO-1; BL-4,6	a) b)	Write a C++ program to overload the triangle function to check whether a Triangle is Equilateral, Isosceles or Scalene. Difference between call by address and call by reference (Three points). Add two numbers using call by reference.	[10] [10]
Q4 CO-3; SO-1; BL-6	a) b)	Write a C++ program to find greatest number among two numbers from two different classes using friend function. Write C++ program for following given information: 1. Class cricketer : cricketer_name, age,team,total_matches_played 2. Class bowler: wickets_taken, bowling_avg 3. Class batsman: run_score,batting_avg Input() and display() are the member functions of the class. Write a program for the given scenario.  	[10] [10]
Q5 CO-2,4; SO-1,2; BL-3,6	a) b)	Draw a Class Diagram for Hospital Management System Write a C++ class for player object with the following attributes player no., name, number of matches and number of goals done in each match. The number of matches varies for each player. Write default constructor which initializes player no., name, number of goals and destroy the constructor at the end by displaying message as "Constructor is destroyed".	[10] [10]
Q6 CO-2; SO-1; BL-6	a)	Class internal_marks having data members are name, roll_no, OOP_marks, SE_Marks and Class external_marks having data members are name, roll_no, OOP_marks, SE_Marks, TCS_Marks. Perfrom addition between internal (out of 50) and external marks (out of 50) of OOP_marks,	[10]

		SE_Marks , TCS_Marks and display the final marks of both the classes using friend class concept.	
	b)	Write four different cases of default arguments. Write Single sum function to add 2 numbers, 3 numbers, and 4 numbers by using the concept of default argument.	[10]
Q7 CO2-; SO-1; BL-6	a)	WAP to perform unary increment and decrement operator using member function.	[10]
	b)	Write a C++ program to design a base class Person (name, address, phone_no). Derive a class Employee (eno, ename) from Person. Derive a class Manager (designation, department name, basic-salary) from Employee. Write a C++ program for Accept all details of 'n' managers. and display all the information of the base classes into the manager class.	[10]